FACTORING MORPHISMS THROUGH THE CONE OF A NULL-HOMOTOPIC COMPOSITE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2772-2788, following the statement at 2761-2770.

Editorial note. No omissions: both factorizations are displayed applications of lemma-functorial-cone.

Allowed standard prerequisites (editorial).

- Functoriality of the cone. lemma-functorial-cone (same chapter, cited at ~2778 and ~2784; queued for the next batch). Produces both factorizations.

Source reference. Original lemma 014F: sources/stacks-derived/pinned-original-source.tex, source lines 2718–2718. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Suppose that $f : K^\bullet \to L^\bullet$ and $g : L^\bullet \to M^\bullet$ are morphisms of complexes such that $g \circ f$ is homotopic to zero. Then*

(1) *g factors through a morphism $C(f)^\bullet \to M^\bullet$, and*
(2) *f factors through a morphism $K^\bullet \to C(g)^\bullet[-1]$.*

Proof. The assumptions say that the diagram

$$\begin{array}{ccc} K^\bullet & \xrightarrow{f} & L^\bullet \\ \downarrow & & \downarrow g \\ 0 & \longrightarrow & M^\bullet \end{array}$$

commutes up to homotopy. Since the cone on $0 \to M^\bullet$ is $M^\bullet$ the map $C(f)^\bullet \to C(0 \to M^\bullet) = M^\bullet$ of Lemma 014F is the map in (1). The cone on $K^\bullet \to 0$ is $K^\bullet[1]$ and applying Lemma 014F gives a map $K^\bullet[1] \to C(g)^\bullet$. Applying $[-1]$ we obtain the map in (2). □

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